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Causes and consequences of forest cover transformation on human-wildlife interaction in the Teknaf Wildlife Sanctuary, Bangladesh☆

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ABSTRACT

Protected Areas (PAs) in Bangladesh have been pivotal for conserving wildlife. The country plans to increase the number of PAs by implementing new forest-governing policies with the involvement of local communities. Their efficacy in wildlife conservation and forest preservation is being questioned due to significant land cover changes within PA boundaries. It is essential to understand the current role of existing PAs in forest cover transformation and their impact on wildlife conservation before expanding the PAs and implementing new policies. This study aims to explore the land cover transformation within Teknaf Wildlife Sanctuary (TWS) and describe the role of the local community and its impact on wildlife. Utilizing Landsat satellite images from 1989 to 2015, this study discovered a 46 % deforestation rate within TWS, surpassing its surrounding areas. Since 1989, 75 % of primary forests have been eradicated, leaving a mere 650 hectares intact. This transformation is shifting forested areas towards the PA boundaries, potentially threatening conservation efforts by making wildlife more vulnerable. Socioeconomic surveys of 5769 households revealed that bigger landowners within the local communities played a significant role in this deforestation. Such community-driven forest alterations increased wildlife vulnerability by intensifying their interaction with forest surrounding communities. This study presents an interesting case of the process of forest transformation within PAs with the intersection of human-wildlife conflict and deforestation. For effective wildlife conservation, strategies need to address and rectify these forest cover transformations and focus on the wildlife habitats, ensuring conservation and coexistence for forests, humans and wildlife.

Introduction

Ensuring the coexistence and prosperity of human communities living alongside wildlife-protected areas is crucial for sustainable development (Andrade and Rhodes, 2012; Braczkowski et al., 2023). Wildlife often kills livestock, destroys crops, and threatens human life. However, they play a crucial role in managing ecosystem balance (O'Bryan et al., 2018), and their sharp decline demands immediate conservation efforts. Nearly 1.60 billion people live within 5 km of forests and are dependent on forests for their livelihoods (Newton et al., 2020). Human-wildlife interaction is inevitable and critical for a significant proportion of the global population. The interaction is varied and complex (Hill, 2021). Generally, when the interaction has a positive (or neutral) outcome it is coexistence and when negative, it is conflict. For the conservation of wildlife respective to human interaction, a balance between minimum conflict and maximum coexistence is required. This balance depends highly on the capacity of the forests to support the wildlife and the dynamics between people-forest relationships.

Establishing and maintaining protected areas (PAs) are effective means of preserving forests and are considered as the spearhead to conserve global biodiversity (Watson et al., 2014). In 2020, 258,608 PAs were designated globally, covering 20.28 million km^2 or nearly 15 % of the Earth' s land surface (UNEP-WCMC, 2020). These PAs host most of the remaining wildlife, and the conservation of wildlife highly depends on the preservation of PAs. Due to the global population increase and development pressure, PAs have expanded conceptually from a

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centralized to a community-oriented approach, with the aim to provide benefits to the surrounding community while conserving wildlife and biodiversity (Lemos and Agrawal, 2006;Watson et al., 2014). Even with the expansion and highly followed means of conserving biodiversity in many countries, the effectiveness and success of PAs have raised serious concern (Coad et al., 2019; Joppa et al., 2008). Studies found that the existing PA network is extremely unlikely to protect the world's mammals (Williams et al., 2022). In the case of terrestrial wildlife, deforestation within PAs is a major concern. It can lead to ineffective conservation efforts by increasing competition and conflict between humans and wildlife. This issue requires serious global and regional level attention and more research needs to be done to understand the how the wildlife interaction is affected by the forest transformation.

Bangladesh is a densely populated country in South Asia, with one of the lowest per capita forest land (0.012 hectare/person) and only 14 % forest cover, where the global per capita forest is 0.52 hectares and the coverage is more than 30 % (FAO, 2020). Moreover, deforestation is a serious issue where forest loss occurred at a rate of 106 km^2 per year from 2006 to 2014, with a total loss of nearly 40 % of the country's entire forest area since 1930 (Reddy et al., 2016). Natural forests in Bangladesh are state-owned and managed by the Bangladesh Forest Department (BFD). Like other countries, Establishing PAs is the leading approach of conserving natural forests and wild habitats in Bangladesh and currently there are 60 PAs, among which, more than half of them has been established after 2000 (BFD, 2021). During the early 1990s community-oriented forest management approaches gained popularity to govern the PAs, especially in the developing nations (Speer, 2012). In case of Bangladesh, forest co-management (co-management is the generic term for community forestry used by BFD) started in 2004 with 5 PAs under a project known as Nishorgo Support Project or NSP (BFD, 2020). Consequently, the model of co-management implemented during NSP were replicated to more PAs (Manzoor Rashid et al., 2013; Mukul et al., 2017; Mukul and Manzoor Rashid, 2017; Mukul and Quazi, 2009) and currently there are 21 PAs under co-management approach (BFD, 2020). The current PA coverage is 23 % (BFD, 2020) and BFD has planned to expand the PA coverage to 30 % of the country's forest land and bring more PAs under co-management approach (BFD, 2016).

PAs host most of the wildlife including large mammals in Bangladesh. Adaptation of new governance policies and its implication on forest cover change is critical for the wildlife conservation. PAs in Bangladesh have concerns over the conservation outcomes and even experiencing deforestation inside their boundaries (Rahman and Islam, 2021). Even though the number of PAs increased recently and the co-management approach was adopted in many PAs, the success of PAs is still being determined. Besides Rahman and Islam (2021), several other recent studies also reported forest loss and deforestation inside PAs in Bangladesh even after implementing co-management approaches (Islam et al., 2016; M.W. Islam et al., 2018, 2019a; Moslehuddin et al., 2017; Rahman et al., 2016). Moreover, it was reported that PAs under co-management had lesser biodiversity status than overall PAs (Islam et al., 2020). Besides forest and biodiversity loss, Islam et al. (2018) reported low level transparency and accountability in the decision-making process, Rahman et al., (2012) found poor governance system and Islam et al., (2021) reported a negative perception regarding conservation outcomes in PAs under the co-management approach. Mollick et al., (2021) pointed out that the co-management approach for PAs in Bangladesh had a wide implantation gap between the ambitious policy framework and the actual rollout. Despite some positive outcomes reported by sevral studies (Alam et al., 2015; Rahman and Islam, 2021; Rahman et al., 2017), PAs in Bangladesh require more initiatives from the forest managers and related stakeholders, even after the latest implementation of the co-management approach. It seems that, the establishment of PAs in Bangladesh has outpaced evaluation and monitoring efforts, where an increase in quantity (coverage and number of PAs) is prioritized over maintaining quality, regardless of the impact and consequences on wildlife and forest conservation. When PAs fail to conserve forest cover, the interaction of wildlife and surrounding communities often turns out to be conflicting due to the dependency on the same forest resources (Billah et al., 2021b; Sarker and Røskaft, 2010). To reduce the conflicting nature of human-wildlife interaction in PAs of Bangladesh, it is important to understand the process of land cover change within the PAs and explore the responsible drivers.

This study focused on a PA known as the Teknaf Wildlife Sanctuary (TWS) in the southern region of Bangladesh, where the land cover change is elucidated with a particular interest in forest cover transformation and its impact on wildlife interaction (Fig. 1). The TWS has undergone continuous deforestation during the implementation of various conservation initiatives over the last century (Tani and Rahman, 2017). Several studies have determined the land cover change, deforestation and forest fragmentation within the boundaries of different PAs (Rahman and Islam, 2021; Billah et al., 2021a; Zafar et al., 2021 and Tani, 2018). But these studies were concerned on the forest cover change or deforestation between a time period. As there are various tree plantation initiatives taken by BFD in and around the PAs, the destruction of natural forest is often not specified in these temporal land cover analyses. Even in TWS, Tani and Rahman (2017) reported tree plantation plots and homestead gardens inside the reserve forests. Due to plantation initiatives, the state of natural forests is often masked under the increased tree coverage from planted trees. But natural forests are irreplaceable because they provide more biodiversity service than any other form of tree coverage (Gibson et al., 2011). Tree plantation can increase the forest coverage, sequester carbon and conserve soil but have little impact on biodiversity conservation (Rudel et al., 2005). TWS is also facing a similar situation where decrease in natural trees is making it hard for wildlife to sustain. TWS has a record of human-wildlife interactions from recent times to a distance past (Hossen and Røskaft, 2023; Islam et al., 2011). Now it is important to understand how changing forests are impacting the human-wildlife interactions.

Besides the forest transformation, it is essential to understand the responsible drivers. In the case of TWS, several studies identified various drivers including betel leaf (*Piper betle*) cultivation (Sakamoto and Tani, 2013; Tani, 2017a; Ullah et al., 2020), fuelwood extraction (Asik and Masakazu, 2017; Ullah and Tsuchiya, 2017) and settlement encroachment (Tani, 2017b; Tani et al., 2013, 2011). But to understand the deforestation process properly, it is necessary to know the extent of involvement of the local community and the socioeconomic dimensions associated with the deforestation process. Therefore, the objectives of this study are 1) to determine the land cover change in TWS, 2) to elucidate the forest transformation to understand the state of natural forests over the time period, 3) to determine the extent of involvement of the local community in various deforestation drivers alongside their socioeconomic status and 4) to explore the nature of wildlife-human interaction due to the change of forest cover. PAs in Bangladesh needs to operate more effectively to conserve the wildlife besides their geographical expansion and adaptation to co-management approach. This study generates a better understanding of PAs' impact in conserving wildlife by addressing the issue of natural forest destruction and human-wildlife conflict. Moreover, the community involvement behind this forest transformation will be useful for understanding the reasons behind this transformation. This new knowledge will also be useful for developing effective management strategies and suitable forest policies in newly established PAs as well as the existing ones and ensure the conservation of wildlife and its habitat.

Limitations and specification

The study focuses on forest transformation from 1989 to 2015, complemented by 2015–2016 community data related to forests dependence. While this may seem dated, the selection is deliberate for two pivotal reasons: a) Pre-Refugee Influx Analysis: The region of TWS underwent a substantial refugee influx in 2017 and currently hosting

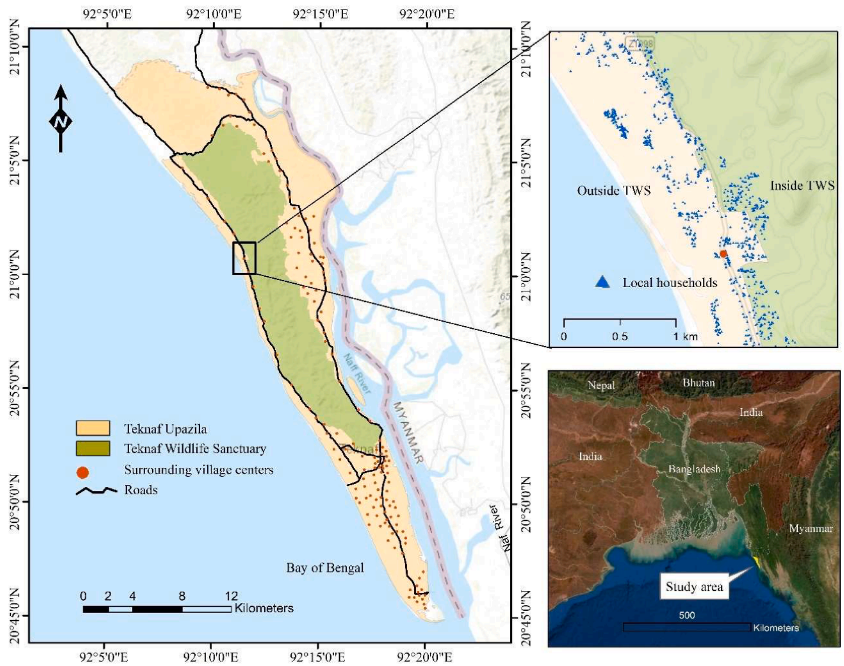

Fig. 1. Map of Teknaf upazila showing the encroached households around the TWS.

about a million refugees. This influx led to accelerated deforestation and wildlife conflict which are distinct from typical deforestation due to community pressures or subpar management. Handful of studies have covered the refugeed induced biodiversity and environmental crisis of the region (Hassan et al., 2018a; Imtiaz, 2018; Rashid et al., 2021; Sarkar et al., 2023b, 2023a). Thus, this research purposefully examines the period preceding this influx to make the results of the study relatable to regional and national scale and imply while designing future forest management policies. b) Synchronizing Remote Sensing and Community Data: The core of this study is a comprehensive 2015–2016 community survey encompassing 147 villages and over 5000 households, shedding light on community-driven deforestation dynamics. To maintain consistency, a corresponding period was chosen for remote sensing analysis. The combination of long-term remote sensing data with large-scale social data covering the entire locality will enable to produce deeper insight to the people, deforestation and wildlife-interaction nexus, not just specific to TWS, but also invaluable for shaping future forest policies in similar community-integrated protected areas.

Methods

Study area

Teknaf Upazila under Cox's Bazar district is the concerned study area where TWS is situated. It is a PA covering 11,615 ha and has a history of rich biodiversity. The boundaries of TWS was identified and extracted from Mouza maps provided by the Land Record and Survey Department of Bangladesh (Ministry of Land, 2020). TWS is among the few forests in Bangladesh where large mammals still occur in the wild (Alam et al., 2013; Nishorgo, 2018). Among the large mammals globally threaten flagship species Asian elephants (*Elephas maximus*) are the most famous. Although their habitat has been severely damaged and fragmented putting a serious concern over the remaining population. State recognition and governance of the TWS began in 1907 as a reserve forest, after which the status changed to the Teknaf Game Reserve (TGR) in 1983; the co-management approach was implemented in the TGR in 2004, and the region finally became a wildlife sanctuary in 2010 and was renamed as the TWS (Khan et al., 2009; Tani and Rahman, 2017 and BFD, 2020).

TWS was under every forest management and conservation initiatives taken by BFD in the forms of mega projects since 2004 and yet experienced deforestation. This makes TWS an important PA to study deforestation and forest conservation. TWS is surrounded by local settlements and these local settlements are linked with the deforestation process (Tani and Rahman, 2017). More than 57,000 households are spread across 147 villages in Teknaf upazila (Ullah, 2018). Upazila is an administrative unit under district administration and is divided into five unions: Bharchhara, Teknaf Sadar, Nhila, Whykhong, and Sabrang. In addition to the TWS, this study also conducted surveys in these unions to understand the socioeconomic profile of the local communities.

Remote sensing data for determining forest land cover change

Remote sensing is the dominant technique for obtaining information on any object or area from a distance, typically from aircraft or satellites (NOAA, 2020). Recently, the number of Earth observation satellites, image quality, and accessibility has increased by several fold, making satellite imagery the most consistent and effective remote sensing tool for large-scale forest monitoring (Belward and Skøien, 2015). Land Remote Sensing Satellite (Landsat) is widely used to investigate forest cover change because of its high spatial resolution and unparalleled temporal depth, which provides the Earth's longest satellite data record (Hadi et al., 2018). In our study, we used Landsat images of 1989 and 2015. From the early 1980s to 2015, similar class Landsat images are available.

The considered Landsat images for this study consist of 30 × 30-meter cells. Each cell is assigned with values measured by multiple sensors. Landsat images cells within the study area were processed and categorized according to the values of normalized differential vegetation index (NDVI). NDVI is an index of the density of green plants on a patch of land and is calculated based on the difference in the values between the visible red and the near-infrared sensors. Its value may range between −0.1 and 1.0, with 1.0 indicating most active green. NDVI has been extensively applied in forest-related analyses based on remote sensing data (Rouse et al., 1974). First, Landsat cells that have less than 0.4 NDVI were eliminated from the analysis because these places are not probably covered with much vegetation. The remaining cells were classified by changes in NDVI during dry season of a particular year. For this study, cells were clustered using the Iterative Self-Organizing Data

Analysis Technique (ISODATA) based on a series of NDVI values obtained for every month from the beginning of the dry season (November) to its conclusion (April) during any 1 year considering the cloud coverage below 5 % . As a result 3 classes of cells were identified, of which Class 3 cells, which entitled the highest NDVI value, were postulated to contain mature trees or forest.

Ground truth data collected at 63 locations classified as class 3 indicated that these cells did not necessarily contain mature trees. This is because the presence of grassy plants and shrubs could also result in a high NDVI value. Therefore, a second step in the analysis conducted on Class 3 cells entailed segregating these cells into those with trees and those with only few trees. For this step, one satellite image for a year was selected for analysis. The timing of the image was as close as possible to the end of the dry season, when the majority of grassy plants were likely to have died, so that the NDVI value reflected the presence of perennial trees. Each cell belonging to Class 3 within this selected image was analyzed, focusing on three types of spectral characteristic: NDVI, Normalized Difference Water Index (NDWI), and Green-Red Vegetation Index (GRVI). This analysis yielded three new clusters (3,4 and 5) produced out of former Class 3, resulting in a total five classes. The presence of trees within the locations of the three new classes was verified using high -resolution satellite images and the ground truth method, confirming that cells within Classes 4 and 5 had trees.

The remote sensing methods for this region considering TWS is described in detail by (Ullah et al., 2022). This study used Landsat imagery and the normalized differential vegetation index (NDVI) to gauge vegetation density. Cells with NDVI below 0.4 were discarded, and the remaining cells were grouped using the ISODATA method, identifying three classes. Ground data showed that high NDVI could also mean grasses or shrubs, not just trees. Therefore, a follow-up analysis utilized NDVI, NDWI, and GRVI to refine Class 3 into three new classes. High-resolution imagery and ground checks confirmed trees were present in Classes 4 and 5. Finally there were 5 land cover classes which were determined as Forest, Lean Forest, Mixed, Bush and Grass.

In this study raster images from 1989 to 2015 containing 5 classes each were considered to elucidate the forest transformation. The raster images were visualized using ArcMap 10.8.1 to produce land cover change maps under UTM 46 N coordinate system (Fig. 2). A raster calculator function from ArcMap 10.8.1 was used to detect the changes in raster pixels from 1989 to 2015 to determine the land cover transformation. The raster images covered the available areas where clear satellite images were obtained from 1989 to 2015. After excluding the beaches, the study area for forest transformation was 20,577 ha, as presented in Fig. 2. The extracted TWS boundary from mouza maps were converted into a GIS layer. The area inside this GIS layer was considered as inside TWS and the remaining study area outside the TWS was referred as around or outside TWS (Fig. 2).

Validation of land cover classes and their transformation

Five land cover classes were generated using Landsat images, and the classes associated with higher NDVI values indicated forest areas and the lower NDVI value indicated grass lands. Based on NDVI values, juxtaposing on google earth images and knowledge of the landscape the classes were considered as the following land cover classes – class 1 (grass), class 2 (bush), class 3 (mixed), class 4 (lean forest) and class 5 (forest). Further the land cover classes were validated using ground photos obtained from transect walks in the TWS during November 2016. A total of 515 ground photos obtained using a GPS-enabled GoPro camera were considered, and the photos were juxtaposed on the 2015 raster image by re-projecting on the UTM 46 N coordinate system. These photos were categorized into six groups based on their vegetation attributes: grassland with small bushes, short bush (waist height), tall bush (above waist heights), bush area with low tree coverage (<10 % canopy coverage), lean forest (10–40 % canopy coverage), and forest (<40 % canopy coverage). Ground pictures from each group are presented in Table 1. Based on visual inspection and NDVI values of the pixels on raster images, the land cover classes were defined as grass, bush, mixed, lean forest, and forest. Forests had mature trees with moderate canopy coverage, and lean forests had fewer trees than forests with low canopy coverage. The mixed area had very few trees with tall and short bushes. Bush mainly included short and tall bushes with no trees. Grass included

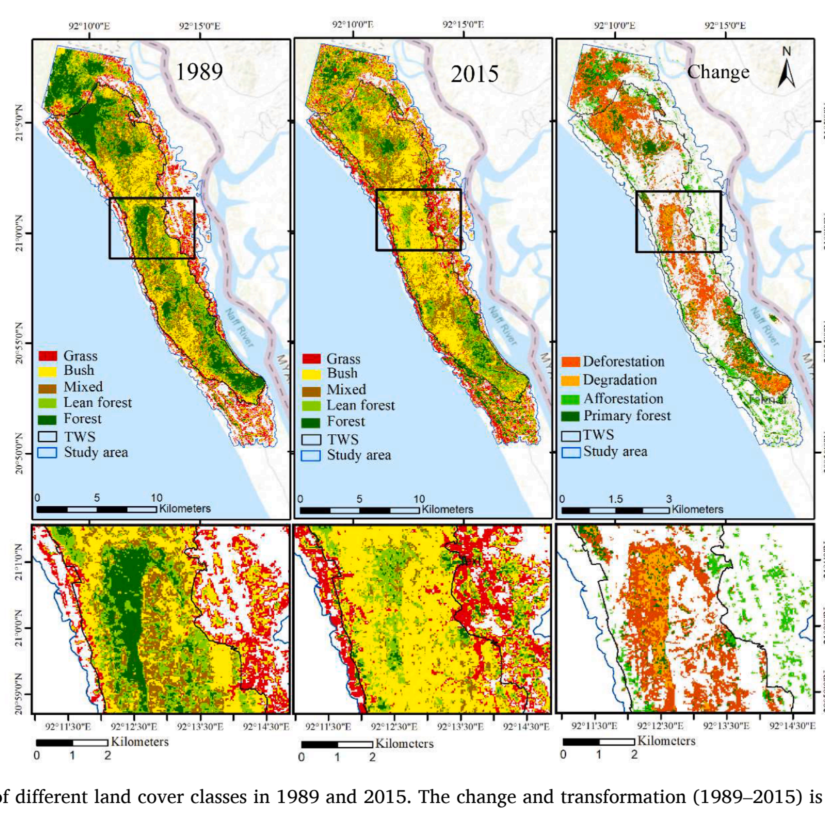

Fig. 2. Land cover map of different land cover classes in 1989 and 2015. The change and transformation (1989–2015) is presented in the right side.

Table 1
Land cover class validation and definition from ground images. Ground photos were compared against the raster cells of the land cover classes. Based on the distribution of ground photos among various raster cells, the land cover groups were determined.

Land cover classes from satellite images	% of ground photo by different categories						Total number of photos
	Grassland with small bushes	*Short bush*	*Tall bushes*	*Bush area with low tree coverage*	*Lean forest*	*Forest*	
Grass	50 %	30 %	13 %	7 %	–	–	***24***
Bush	22 %	31 %	36 %	11 %	–	–	***250***
Mixed	16 %	8 %	20 %	40 %	16 %	–	***80***
Lean forest	7 %	7 %	7 %	6 %	67 %	6 %	***65***
Forest	–	–	–	–	3 %	97 %	***96***

cultivable lands, hilly grasslands, and small bushes. The definition and validation process is described in detail by (Ullah, 2018).

To adequately describe forest changes and transformations, the relevant terminologies must first be defined. In this study, land cover refers to the remotely-sensed physical material at the surface of the Earth, such as vegetation or open water (Noordwijk et al., 2016). The land cover of the study area was categorized into the five classes. Forests can be defined by various terms based on vegetation attributes, legal status, or even land use. Here, forest land was defined as the area of land proclaimed to be forest under a Forest Act or Ordinance, regardless of whether the area had tree cover (Ford-Robertson, 1971). In this study, forest land refers to the TWS, and forest refers to the raster pixels termed as forests in the land cover classification.

Land cover change refers to the increase or decrease in the coverage of a certain land cover class, and transformation refers to the conversion of one land cover class to another. Land transformation was determined by the conversion of land cover classes presented in Table 2. The definition of the transformation of deforestation, degradation, and afforestation was adopted from Lund (Lund, 2018). Deforestation is the conversion of forest and lean forests to other land cover classes, degradation is the conversion of forests to lean forests, and afforestation is the conversion of other classes to forest or lean forests. Forests that remained unchanged during the study period were defined as primary forests, and the transformation of other land classes to forests was defined as secondary forests.

Socioeconomic survey and sample of the local community

Primary data were collected through a household survey conducted from December 2015 to May 2016. Household heads were interviewed using a structured questionnaire. During the interview, questions related to family size, education, annual income, and land property were asked. The entire population (except residents of St. Martin, a small island near the coastline) of Teknaf upazila was considered for this study. During the survey, we covered all 147 villages and listed all the households with their location data. In addition, we conducted an interview survey covering 10 % of all households. To select the 10 % sample households for interviewing, we followed a systematic sampling procedure where every 10th household was interviewed during the listing of all households in the community. If any household had no members able to respond, we moved to the next household for interviewing and from there again every 10th next household was interviewed. Ultimately, 57,404 households were listed, of which 5769 were interviewed to understand the socioeconomic profile of the local community. The socioeconomic survey followed the methodology described by (Angelsen et al., 2014) for measuring livelihoods and environmental dependence. As the survey covered a large population size, local survey assistants (6–10) were recruited to conduct the survey after undergoing thorough training from the research team. Households were grouped based on their ties to deforestation drivers: betel leaf cultivation, fuelwood collection, and encroachment within the TWS. Those cultivating betel leaf were linked to that activity. Households collecting fuelwood from the restricted TWS or located inside the TWS were classified accordingly. These classifications were drawn from household interviews and location data from the survey. Household-level data was aggregated to village-level data, and the relationship between engagement with deforestation drivers and economic condition was explored. Interview data included the household head's age, education, primary occupation, family size, residency duration, property size, and yearly income. The data was analyzed using both descriptive and inferential statistics in SPSS version 16, focusing on the community's socioeconomic aspects.

Human-wildlife interaction data

While conducting the household survey, every villages (147) were covered and informal chats regarding wildlife gave an idea about the

Table 2
Land cover and forest transformation.

Land cover class	**Land transformation**
Forest	Deforestation
Lean forest	Degradation
Mixed	Afforestation
Bush	Primary Forest
Grass	Secondary forest

pattern of wildlife interactions. The author also had encounter with wild animals and took picture and documented it. Besides elephant dung was frequently found near the forest areas. Three Focus Group Discussions (FGDs) containing 12~15 participants each were conducted in Baharchhara, Teknaf Sadar and Nhila union. The FGDs covered topics including deforestation, forest dependency and wildlife interactions. Discussions were led by moderators using prepared checklists and also designating free form of discussion by the participants. The information elicited included the concern for wild animals, the changed interaction patterns overtime and resource depletion inside the forests.

Results

Land cover change and transformation

Land cover change and transformation from 1989 to 2015 in the Teknaf Peninsula are presented in Fig. 2. Here the reduction of forest coverage (dark green pixels) over the considered study period is illustrated. The amount of change in hectares is shown in Table 3. From 1989 to 2015, substantial changes in land cover occurred in the forest and bush areas. Forest area decreased by 46 % from 3734 ha to 1974 ha. The bush area increased by 49 % from 4624 ha to 6899 ha. Therefore, the dominant land cover transformation was the conversion of forest land to bushes. Among other land cover types, grass and mixed areas increased by 17 % and 12 %, respectively.

Although forest area exhibited a steep decline, the lean forest area increased by 9 %. Table 3 presents the land cover transformation. From 1989 to 2015, only 925 ha of forest remained untouched and was therefore referred to as primary forest. This area of primary forest was the only intact natural forest of the original 3734 ha forested area in 1989. The total area of deforestation (considering both forest and lean forests) was 3378 ha, and the total area of forest degradation was 1389 ha. In contrast, 1939 ha were afforested, of which 1048 ha were secondary forests. Moreover, 7955 ha of non-forested areas were transformed, whereby land cover interchanged between grass, bush, and mixed areas.

Forest transformation inside and around the TWS

In total, the forest area decreased by 46 % or 1760 ha from 1989 to 2015. When not considering the secondary forests, the actual decrease was 2809 ha, which is nearly 75 % of the original 3734 ha of natural forest in 1989. To summarize the total forest transformation, 2809 ha of primary forest were destroyed, and 1049 ha of secondary forest were afforested, resulting in a total forest area decrease from 3734 ha to 1974 ha. Decreasing forest area in a PA zone highlights major insufficiencies in the current conservation approach. Moreover, forests in PAs are expected to be untouched or convert at slower rates than that of unprotected areas.

Our results suggest that forests have been decreasing around and within the TWS region. Fig. 3 shows the forest transformation both inside and outside the TWS. The TWS comprised 2727 ha of forests in 1989, which decreased by 61 % to 1049 ha forests in 2015. In contrast, the forest area outside the TWS decreased by only 8 % from 1007 ha to 925 ha during the same time period. Fig. 3 shows that deforestation and degradation rates inside the TWS were significantly higher than afforestation rates. However, deforestation and degradation rates outside the TWS were almost equal to afforestation rates. The actual decrease in forest area (the calculated decrease excluding afforestation) inside and outside the TWS was 76 % and 73 %, respectively. The total deforestation was higher inside the TWS than outside (61 % compared to only 8 %) due to the much lower afforestation rates inside the TWS.

The TWS comprised 1049 ha of forests in 2015, among which only 38 % were afforested. In contrast, 70 % of the forests outside the TWS were afforested. Fig. 3 shows that the conversion of forest to other land classes was greater than the conversion of other land classes to forest areas inside the TWS in 2015. The conversions were approximately equal outside the TWS. Fewer areas inside the TWS were converted from bush or grassland to forests in 2015 compared with those outside. More specifically, 642 ha of forests from 1989 were converted to bush and grassland, but only 225 ha of bush and grassland were converted to forests in 2015. In contrast, the conversion of bush and grassland to forests and vice versa was almost equal (slightly higher in terms of afforestation) outside the TWS. In addition to deforestation inside the TWS, nearly one-third of the forest area in 1989 degraded to lean forests in 2015. Only 650 ha of primary forest remained inside the TWS in 2015, which is only 24 % of the total forest area in 1989. To summarize the forest change, deforestation and degradation occurred both inside and outside the TWS but deforestation and degradation were much greater than afforestation inside the TWS. This resulted in a higher total decrease of forest inside area than that outside. Moreover, when considering the transformation of forests, approximately one-fourth of primary forests remained inside the TWS in 2015, with fewer areas of secondary forests. More secondary forests were observed outside the TWS due to afforestation efforts from the BFD.

When comparing the nature of land cover transformation between the inside and outside of the TWS, deforestation and afforestation were more disproportionate than previously assumed. Fig. 4 compares the% forest transformation between inside and outside the TWS. We found that 77 % and 82 % of the deforestation and degradation in the region occurred inside the TWS, respectively. In contrast, only 41 % of afforestation and 38 % of the secondary forests occurred inside the TWS. Therefore, deforestation and degradation were greater, and afforestation was lower inside the TWS than outside. As TWS was comprised of natural forests in the past, 70 % of the remaining primary forests were still found inside the TWS. The higher deforestation rate inside coupled with the higher afforestation rate outside will inevitably shift the tree coverage from the core forest land to the surrounding locality. The forest transformation trend suggests that the natural trees inside the forest are being harvested, and local communities are planting more homestead trees for their own consumption.

Therefore, this study identified fragmented patches of primary forests in the core region of the TWS with very few secondary forests. Most of the secondary forests occur near and around the TWS boundary, while primary forests are scarcely observed in the boundary region. Fig. 5 depicts the location of the primary and secondary forests inside the TWS. Primary forests rarely occur within 250– 500 m of the TWS boundary, and few patches are observed within 500–1000 m of the TWS boundary. Moreover, most of the remaining primary forests occurred as fragmented patches in the core region (location: 21°4'0"–21°6'0" N and 92°10'0"–92°12'0" E; 20°53'0"–20°58'0" N and 92°14'0"–92°16'0" E). Secondary forests were mostly located within 250 m of the forest boundary and scarcely within 250–500 m.

The main cause for the regenerated secondary forest was the encroachment of homestead gardens and tree plantation plots via the social forestry program (Asahiro, 2018 and Asahiro et al., 2011). The remaining few secondary forests were scattered in the TWS without

Table 3
Land cover change and transformation from 1989 to 2015. Land cover change is shown according to the percentage of area coverage change of different land classes during the considered time period. Land transformation is stated based on the area coverage change among different land cover classes with special focus on forests.

Land cover type	Area (ha)		% Change	Land cover transformation	Area (ha)
	1989	2015			
Barren	4144	2870	−31 %	Non-forest area	7955
Grass	2731	3196	17 %	Deforestation	3378
Bush	4624	6899	49 %	Degradation	1389
Mixed	2216	2482	12 %	Afforestation	1939
Lean forest	3128	3156	9 %	Primary forest	925
Forest	3734	1974	−46 %	Secondary forest	1048

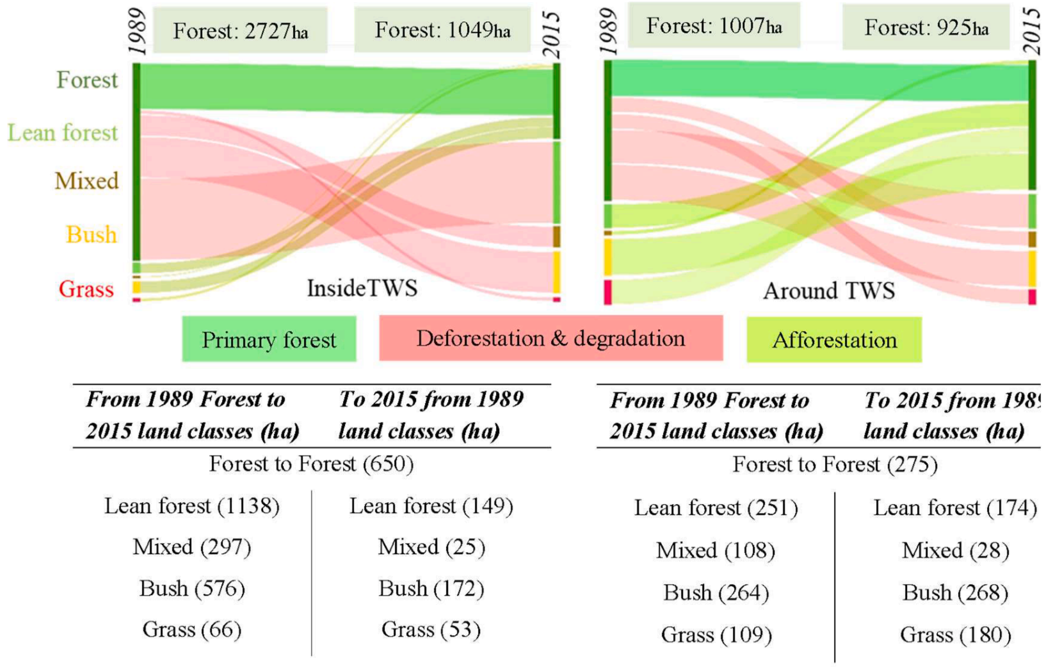

Fig. 3. Forest transformation inside and around the TWS from 1989 to 2015.

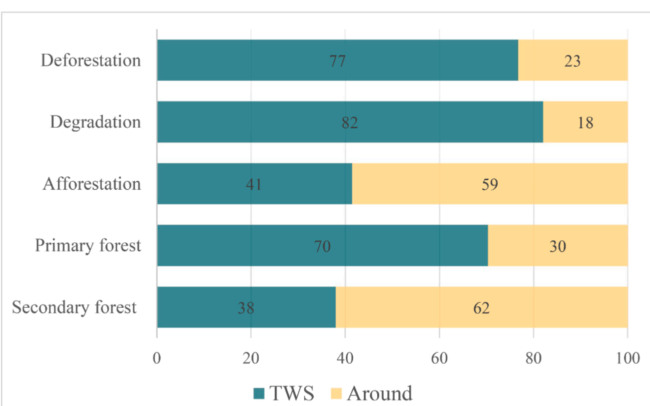

Fig. 4. Comparing the percentage of forest transformation between inside of the TWS and around the TWS.

forming any forest patches. Therefore, from the forest transformation analysis, we conclude that the primary forests are decreasing, with few remaining in the core zone. The secondary forests are regenerating near the boundary region but at a lower rate than the total deforestation in the region.

Wildlife interaction with the community

From the recent to distant past, studies in the region have discussed the various issues related to forest wildlife interaction and conservation, including large mammals, with a special focus on elephants (Billah et al., 2021b; Hossen and Røskaft, 2023; Rahman, 2019). The interactions between wildlife and their potential conflicts within the TWS region have been corroborated by both the findings from the Focused Group Discussions (FGDs) and direct observations by the research team. During the field survey conducted within the local community, the researchers encountered wild monkeys and elephants near the forest periphery on multiple occasions. Fig. 5 elucidates such instances of these interactions. In the TWS region, the presence of large mammals adjacent to the forest boundaries is quite common, as documented by the researchers.

The findings from the FGDs regarding human-wildlife interactions from the TWS region are stated below –

- Wild elephants and monkeys are the dominant large mammal species and often observed proximate to forest boundaries and neighboring homestead gardens.
- Households proximate to the forest, particularly those cultivating fruit species, act as significant attractants for these mammals. Additionally, social forestry plots near forest perimeters serve as critical foraging and shelter zones for these species.
- A decline in the elephant herd sizes has been noted over time. While locals report perceived reductions in the overall number of elephants, they concurrently observe increased encounters with recurrent individual mammals. This phenomenon suggests that while the overall population might be on the decline, the remaining animals are exploring human-dominated territories more frequently, potentially due to habitat degradation and food scarcity.
- Elephants predominantly exhibit non-aggressive behaviors, resorting to hostility mainly when provoked or threatened. Their primary impact on human settlements is the destruction of crop fields.
- The local community unanimously agrees the steadily declining availability of food and habitat within the core forest regions which is the biggest threat on the remaining wildlife.

Given these observations, the encroachments of wild animals into human territories seem to be a direct consequence of forest resource depletion. These patterns emphasize the imperative need for habitat conservation and the development of sustainable strategies that ensure both the survival of these species and the well-being of local communities.

Socioeconomic status of the households related to deforestation drivers

In this study, we found that 2150 households (37 % of total) collect fuelwood, 623 households (11 % of total) were inside TWS, and 506 households (9 % of total) cultivate betel leaves. These households (HHs) were directly related to the major deforestation drivers (DDs) in the region. Table 4 provides the descriptive statistics of the selected

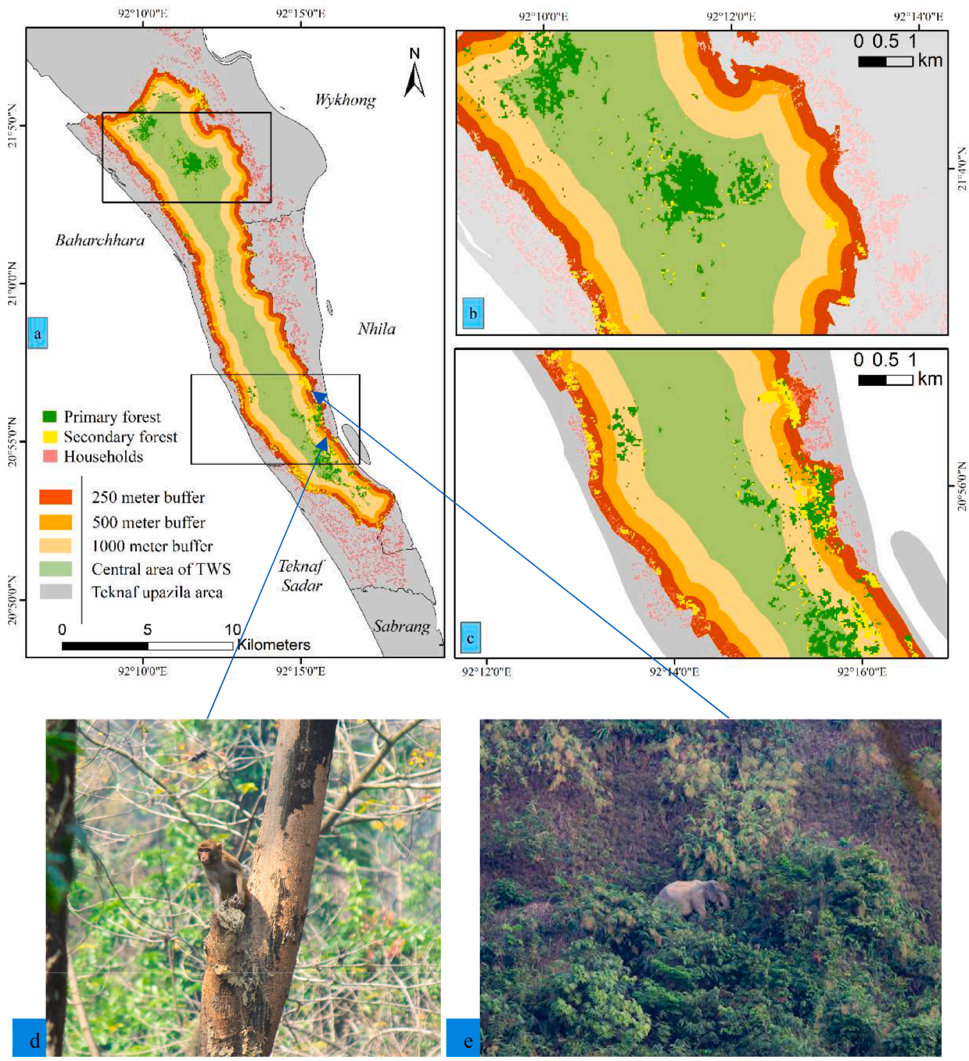

Fig. 5. Primary and secondary forests inside the TWS in 2015: a) Buffer areas of 250, 500, and 1000 m within the TWS boundary surrounded by local households; b) remaining primary forests in the center of the TWS; c) secondary forests near the TWS boundary; d) Long-tailed macaque (*Macaca fascicularis*) spotted by the author in social forestry plots and e) Asian elephant (*Elephas maximus*)spotted by the author within 500 m in the forest boundary.

Table 4
Socioeconomic characteristics of households according to deforestation drivers.

Deforestation driver category	Socioeconomic variables								
		Age (Years)	Tenure (Year)	Family (Members)	Education (Year)	Land (ha)	Income 000BDT)	Poor[1]	Above Poverty[2]
	N	Mean±SD						% of Frequency	
Fuelwood Collection (F)	*2150*	40.75 ±13.01	14.11±19.93	6.14±2.36	1.61±2.94	0.11 ±0.23	169±254	69	23
Encroachment (E)	*623*	39.51 ±12.89	10.58±9.69	6.05±2.20	1.63±2.94	0.15 ±0.27	163±143	68	24
Betel leafCultivation (B)	*506*	44.47 ±12.52	18.07±22.10	6.91±2.41	1.29±2.73	0.23 ±0.40	213±195	52	40
(E) + (F)	*426*	40.29 ±13.32	10.86±9.76	6.04±2.19	1.22±2.31	0.16 ±0.29	148±117	73	20
(F) + (B)	*250*	43.90 ±13.42	19.95±23.75	6.96±2.36	1.29±2.63	0.22 ±0.35	204±219	58	35
(E) + (B)	*123*	42.15 ±13.02	13.97±8.74	6.84±2.23	1.20±2.31	0.28 ±0.39	185±131	62	32
All	*93*	40.78 ±13.52	14.42±9.93	6.68±2.28	1.27±2.38	0.29 ±0.39	170±118	66	28

[1] (HIES, 2018) considered monthly expenditure to determine poverty, and households with an annual income of less than 169,872 BDT (monthly 14,156 for rural areas) were termed as poor,.

[2] (World Bank, 2019) used a daily income of USD 1.9 to determine the poverty line, which was adopted and converted using mean family size to the annual household income of 199,167 BDT; therefore, households with an annual income above 199,167 were considered above the poverty line

socioeconomic variables of the households related to the three major deforestation drivers (DDs) and the possible combination of participation in DDs including HHs associated with all DDs.

Regarding age, the heads of households that cultivate betel leaves were slightly older, with a mean age of 44.47 years (mean age of the other households was ~40 years). The mean tenure for all households in the local community was 13.26 years, with a standard deviation of 17 years. Compared to the mean of the local community, betel leaf cultivating households had a higher tenure of 18.07 years, and households inside TWS had a lower tenure of 10.58 years. Besides, HHs combined with encroachment and fuelwood collection had a comparatively low tenure (10.86) compared to others, while fuelwood collection combined with betel cultivation had a higher tenure (19.95). The average family size of the local community was 6.14, with a standard deviation of 2.42. The family size of the betel leaf cultivation households was slightly higher, with a mean of 6.91 members and the largest family (6.91) was found among the combined HHs of fuelwood collection and betel cultivation. The mean schooling year of the local community was 1.97 years, with a standard deviation of 3.46 years. Among the households related to the various deforestation drivers, betel leaf cultivators combined with encroachment had the lowest schooling years (1.20 years), and households inside TWS had the highest (1.63 years). Comparing with the local community, households associated with deforestation drivers had fewer schooling years.

When considering land property, households encroached inside the forests combined with betel leaf cultivation had the largest land property (0.28 ha) and the fuelwood collectors had the smallest (0.11 ha) land property. The mean annual household income of the local community is BDT 195 thousand. Households related to fuelwood collection, inside TWS, and betel leaf cultivation had an annual income of BDT 169,000, 163,000, and 213,000, respectively. The lowest income (148,000 BDT) was found among the households combined with encroachment and fuelwood collection where 73 % HHs were poor with only 20 % above the poverty line. Considering the total population, 61 % of households in the local community were poor, with only 30 % above the poverty line. In contrast, 52 % of households cultivating betel leaves were poor, with 40 % of households above the poverty line. The households associated with the other two drivers were comparatively poor; nearly 68 % of households were poor and only 24 % of households were above the poverty line.

Fig. 6 presents community level association between DDs with income and land. Village level average income had no association with communities engaging with deforestation activities. The village level land had positive correlations with communities being more engaged with deforestation drivers. Communities having access to more land have greater chances being engaged in deforestation activities.

Discussion

This study showed that, 46 % of the forest cover was destroyed during 1989 to 2015 in the Teknaf region and the percentage of decrease of forest covers were even higher while considering inside the reserve forest boundary only. This raises serious concerns over the effectiveness of the current PA to conserve forests and its habitats. Besides deforestation of PAs in a period, this study also determined the conversion of natural forests to other form of vegetation cover. So far our knowledge, this is the first study concerning a PA in Bangladesh, where natural forest destruction was separately explored. Here, while analyzing forest transformation, forests were differentiated among unchanged natural forests (primary forest) and regenerated forests (secondary forest). Differentiating primary and secondary forests provided better understanding of the deforestation process and further elucidated the impact and consequence of PAs on forest conservation. Previous studies analyzed forest cover transformation in several PAs using remote sensing and reported deforestation inside them (Rahman and Islam, 2021; Billah et al., 2021a; Zafar et al., 2021; Tani, 2018) These studies covered deforestation inside the PAs including Hazarikhail Wildlife Sanctuary, Dhopachari Wildlife Sanctuary, Chunati Wildlife Sanctuary, Fasiakhali Wildlife Sanctuary and TWS. On the contrary, studies have also reported gain in forest cover in Hazarikhali Wildlife Sanctuary, Chunati Wildlife Sanctuary and Lawachara National Park (Chowdhury et al., 2020; Islam et al., 2019 and Islam et al., 2019b). However, no studies considered the destruction of primary forests and the gain in forest cover was reported from several plantation initiatives by BFD. This study found that, the region lost 75 % (from 3734 ha to 925 ha) of its primary forests during 1989 to 2015, where the total deforestation in the region was 46%. Among the remaining 925 ha primary forests, 70 % are inside the TWS but higher rates of deforestation inside the TWS boundaries indicate greater risks of future destruction.

The primary conservation roles of PAs are protecting natural forests. Not only in the TWS but also around the global biodiversity hotspots, destruction of natural forests is a serious concern. The global decline of primary forests is often overlooked or underweight due to the notable increase in tree coverage (Song et al., 2018). The loss of natural forests cannot be compensated by increasing tree coverage through plantation initiatives, even inside the forest boundaries. Primary forests are irreplaceable and have higher biodiversity values than secondary forests or any other forms of tree coverage from various tree plantation initiatives (Gibson et al., 2011). Bangladesh is also facing a similar scenario of forest cover change. From 2000 to 2014, the tree coverage of the country increased by 136,000 ha, but nearly 84,000 ha of trees were lost within natural forest boundaries (Potapov et al., 2017). Therefore, the deforestation conditions of a PA and its surrounding region cannot be completely determined by comparing the forest coverage change between two given time periods. In case of Teknaf, 75 % of natural forest

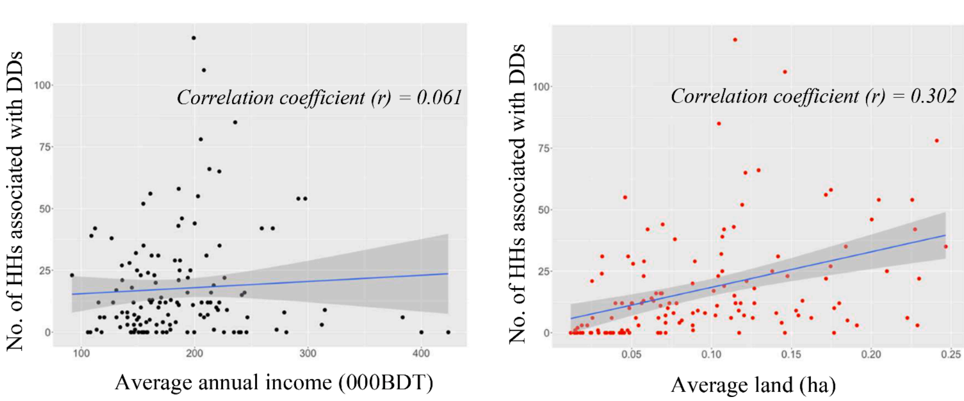

Fig. 6. Community level data showing correlation between village-wise household (HHs) participation in deforestation drivers (DDs) with annual income and land property.

destruction was masked with 46 % total deforestation in the region which may impede to understand the actual crisis in the conservation initiatives.

Besides the destruction of primary forests in TWS, this study also found that the tree coverage was shifting from the center towards the boundary of the reserved forest area. This shifting of forest cover is the consequence of the local community influenced forest transformation process. In the Teknaf region, deforestation and degradation occurred at a higher percentage inside the TWS, whereas afforestation occurred higher outside the TWS. This resulted in, decrease of primary forests in the core region of the TWS and increase of secondary forests near the boundary of the TWS. This study reported that, 77 % and 82 % of the total deforestation and degradation, respectively, occurred inside the TWS and about 60 % afforestation occurred outside the TWS.

The decline of natural forests and shifting of forest cover have major negative impacts on the human-wildlife interaction. In the TWS, the increased interaction between wildlife, notably elephants and monkeys, with the local communities has amplified the vulnerability of these wild species. As habitat degradation and food scarcity drive these animals into human-dominated areas, they face heightened risks, including potential conflict, injury, or even mortality. Simultaneously, these encroachments pose challenges for the local community in terms of safety and crop damage. The sanctuary's reduced core forest resources, coupled with the attractiveness of adjacent agricultural and social forestry plots, underscore a pressing need for adaptive conservation strategies to ensure the sustainable coexistence of wildlife and human inhabitants. Designing buffer zones around the forest area and assigning forestry plots may have positive impacts to provide support to the forest dependent community and persuade them from core forest area but may have devastating consequence on the biodiversity conservation.

Moreover, this study explored the extent of involvement of the local community in the forest transformation process.Geist and Lambin, (2001) identified agriculture, wood harvest and infrastructure development among the proximate drivers responsible for most of the tropical forest destruction. This study reported that among the local community, 37 % of households collected fuelwood, 11 % of households were encroaching inside the TWS, and 9 % cultivated betel leaves using forest resources. Therefore, 57 % of the local community were involved in any form of deforestation drivers. This higher percentage of community involvement indicates the over dependency on the forest resources. Besides the over dependence on forest resources, 11 % households inside the forest boundary and more than half of the community's engagement in various deforestation drivers indicate a serious concern regarding the enforcement of the restrictions inside the PA boundary. It is clear that the strict rules and regulations to protect PAs are not implemented properly in case of the TWS.

The socioeconomic analysis revealed that among the TWS surrounding local community, nearly 61 % of households do not earn enough to meet their monthly expenditure, and 70 % of the population live below the poverty line. Poverty can be linked to forest dependence in multiple and potentially conflicting ways (Nerfa et al., 2020). Impoverished households in forested areas tend to be more dependent on forest products, as they lack alternative resources to supply their needs (Reyes et al., 2018). When forests are over-exploited, forest dependence can become a significant deforestation driver. Previous studies reported several deforestation drivers and their respective impact on forests in TWS. Ullah and Tsuchiya, (2017) showed the high demand of fuelwood, Ullah et al., (2020) explored the impact of betel leaf cultivation and Tani, (2018) reported the impact of encroachment. But this study gave an overall idea about the extent of the local communities involved in these deforestation process. From the results, it is clear that encroached houses with large land properties are playing a significant role in forest transformation.

The policy makers and forest managers need to consider the local communities and their involvement while developing strategies to conserve the forests. Analyzing the related deforestation drivers of the local community is important for determining the conservation impact of PAs. Oldekop et al., (2016) argued that PAs with positive social and economic impacts on the local community tend to have positive outcomes from their conservation efforts. Therefore, the socioeconomic status of the surrounding community can also be an indicator of the PA impacts on forests. As the local community around the TWS is poor and highly dependent on forest resources, focus must be placed on improving the overall economic status of the region, the capability of the local community, and the forest management policies. Deforestation inside PAs make them more vulnerable to natural and manmade calamities. TWS experienced more deforestation while hosting large number of refugee influx after 2017 and the forest habitats are facing serious threat of extinction (Braun et al., 2019; Hasan et al., 2021; Hassan et al., 2018b; Sakamoto et al., 2021). To protect the PAs with its biodiversity and avoid such scenario in the future it is important to stop forest destruction inside the PAs.

Under the existing forest management strategies PAs like TWS are experiencing deforestation with a high rate of primary forest conversion. Tree plantation initiatives may increase the forest coverage but have little impact on biodiversity conservation. Tree plantation initiatives increase carbon sequestration and conserve soil but have little or no impact on biodiversity conservation (Gibson et al., 2011; Rudel et al., 2005). Moreover, such unplanned tree planation near the forest boundary can increase the human-wildlife interaction and result in loss of wild animals. If protecting PAs in Bangladesh continues the planting of tree plots at the perimeter and involving the local communities in the form of co-management as suggested by Ostrom (1990) and Agrawal et al. (2008), some factors must first be considered when amending and implementing forest policies and strategies. Firstly, engaging local communities to manage forests require them to be empowered and capable of taking such responsibilities, without empowerment the participation may be symbolic and non-influential in decision making (Devkota, 2020 and Mustalahti et al., 2020) . Second, planting trees in the PA perimeter can be counterintuitive, as trees may attract more people to the forests from whom conservation initiatives are intending the forest to be protected from (Wittemyer et al., 2008). Third, property rights must be extended beyond a certain limit where the local community have enough access and rights that they feel to protect and manage the forests for their own benefits. Property rights or resource rights limited to "50 % of what you harvest" due to the current co-management rule is ambiguous and not always sufficient to alter deforestation (BFD, 2017). Finally, drastic steps need to be taken to conserve the remaining natural forests and improve forest cover in the core forests with native species.

Based on our findings, we suggest that alternatives to fuelwood, sustainable cultivation practices of betel leaves, and PA zoning based on recent forest cover changes should be prioritized and included within existing forest policies and strategies. While zoning priority should be given to wildlife habitats and zoning must be done keeping in mind of less interaction between wildlife and local communities. Since 2000, almost every country in the South Asian region has increased PA coverage by adopting co-management or other forms of participatory forest management approaches; however, outcomes in biodiversity conservation remain a serious concern (Clark et al., 2013). Therefore, there is no "one size fits all" approach in biodiversity conservation. Instead, regional authorities must implement location-specific policies and strategies based on local empirical findings while also maintaining guidelines from the international community. Special priority should be given to those PAs which are habitat to big mammals.

Conclusion

This study showed that due to the consequence and actions of forest related policy and people the ratio between primary and secondary forests are leading towards an unwanted direction where forests are being degraded and losing the capacity to serve as suitable habitat for wildlife. Consequently the human-wildlife interactions are turning towards a conflicting situation where the possibility of the extinction of large mammals from the forests are being eminent. Primary forests are necessary for supporting the wild habitats and conserving biodiversity which cannot simply be replaced by afforested secondary forests. In the case of the TWS, primary forests are being destroyed from the center of the PA, and secondary forests are being planted in the boundary region. Moreover, most of the secondary forests inside the TWS are the homestead gardens of encroached households with exotic tree species. Therefore, such type of forest cover transformation can result in further biodiversity loss in the PA. To understand the dominant causal mechanisms of such land cover conversion, we identified the extent of involvement of the local community in the deforestation process with their socioeconomic profile. Our study suggested that the high percentage of poverty and over-dependence on forest resources of the local community must be considered by the forest managers and policymakers while developing forest conservation policies.

PAs are the main spearhead for protecting the existing biodiversity and the importance of PAs have been such that scientist are calling for more areas under PAs in the coming days. Protecting the world's existing biodiversity requires well-funded and well-managed PAs covering the most vulnerable places and species, which must also be complemented by global forest conservation policies. Achieving meaningful and effective changes in global policies require both national and regional support. Bangladesh plans to expand its PAs, but a better understanding of the current scenario in existing PAs is required. Besides the expansion, PAs need to be managed more effectively to conserve the biodiversity where the conflicting scenario between the wildlife and local communities must be resolved. Conserving biodiversity in critical PAs surrounding forest-dependent local communities requires evidence-based working plans and strategies supported by empirical research. For the TWS, priority should be to protect the existing primary forests and core forest areas which can promote the future regeneration of native species. This can contribute to improving the capability to support the remaining wildlife and therefore, reduce human-wildlife interactions.

Afforestation programs involving the local community can facilitate greener PAs, enrich their carbon sink, and provide income to the local community, but they are also likely to have little effect on biodiversity protection. The TWS and other critical PAs in Bangladesh must upgrade their forest policies, specific strategies, and detailed working plans that involve the local communities to ensure the strict protection of the remaining natural forests. With limited manpower, inadequate legal authority, and an overly generalized BFD forest policy, the declaration of more restricted forests will likely create "paper parks," and the homestead gardens of households inside the PAs may add new forest covers. Unfortunately, biodiversity loss will be inevitable under these current PA management schemes.

CRediT authorship contribution statement

SM Asik Ullah: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Kazuo Asahiro:** Supervision, Conceptualization. **Masao Moriyama:** Formal analysis, Data curation. **Jun Tsuchiya:** Supervision, Conceptualization. **Md Abiar Rahman:** Supervision, Conceptualization. **Mariyam Mary:** Methodology, Formal analysis. **Masakazu Tani:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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